

Multi-Asset Time-Series Momentum

Risk-aware portfolio construction using liquid ETF proxies

Research report

Daily ETF proxies | 2007-07-02 to 2026-07-01

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Research status This report documents a pre-specified research process. Results are reported only after data validation, validation-period model selection, and a single untouched out-of-sample evaluation.

Executive summary

Table 1: Research brief and current status

Item	Specification
Research question	Do medium-term price trends across liquid asset classes support a diversified, implementable systematic portfolio after transaction costs?
Method	Daily time-series-momentum signals across ten ETF proxies, tested first with equal absolute weights and then with inverse-volatility sizing.
Current status	Data acquisition, integrity checks, and equal-weight validation are complete. No robust lookback was selected after the cost-sensitivity check; inverse-volatility, rebalancing robustness, and test-period evaluation have not been run.
Decision standard	Evidence must remain credible across predefined parameter, cost, and rebalancing robustness checks; a high in-sample Sharpe ratio alone is not sufficient.

1 Research question and hypotheses

1.1 Primary question

Do medium-term price trends contain enough persistent information across liquid asset classes to construct a diversified systematic portfolio with superior risk-adjusted characteristics relative to reasonable passive benchmarks?

1.2 Hypotheses

H_0 . Past medium-term returns contain no economically useful information sufficient to produce a robust portfolio after realistic implementation assumptions.

H_1 . Positive medium-term trends tend to persist and negative medium-term trends tend to

continue sufficiently to support a diversified systematic trend-following portfolio.

Secondary hypothesis. Volatility-aware sizing improves the risk-adjusted characteristics and robustness of the raw momentum strategy.

2 Data

2.1 Universe and source

The analysis uses adjusted daily ETF prices obtained from Yahoo Finance via `yfinance`. The frozen universe covers equities, government rates, credit, real assets, and currency exposure. The aligned sample contains 4,780 common trading dates from 2 July 2007 through 1 July 2026.

Table 2: Frozen research universe

Ticker	Risk source	Role in the universe
SPY	US equities	Broad US large-cap equity beta
EFA	Developed ex-US equities	International developed-market equity exposure
EEM	Emerging equities	Higher-volatility emerging-market equity risk
IEF	Intermediate Treasuries	Moderate-duration US government-bond exposure
TLT	Long Treasuries	Long-duration US government-bond exposure
LQD	Investment-grade credit	Corporate credit-spread risk
GLD	Gold	Precious-metal and real-rate-sensitive exposure
DBC	Broad commodities	Diversified commodity-futures exposure
UUP	US dollar	Long USD currency exposure
VNQ	US real estate	Listed US real-estate exposure

Source: Yahoo Finance, retrieved 10 August 2026.

2.2 Data policy and limitations

Prices are adjusted for splits and distributions. The analysis retains only dates common to all ten instruments; missing values are not forward-filled. The use of ETF proxies, Yahoo Finance data, and a strict shared-history sample are material limitations and remain explicit throughout the report.

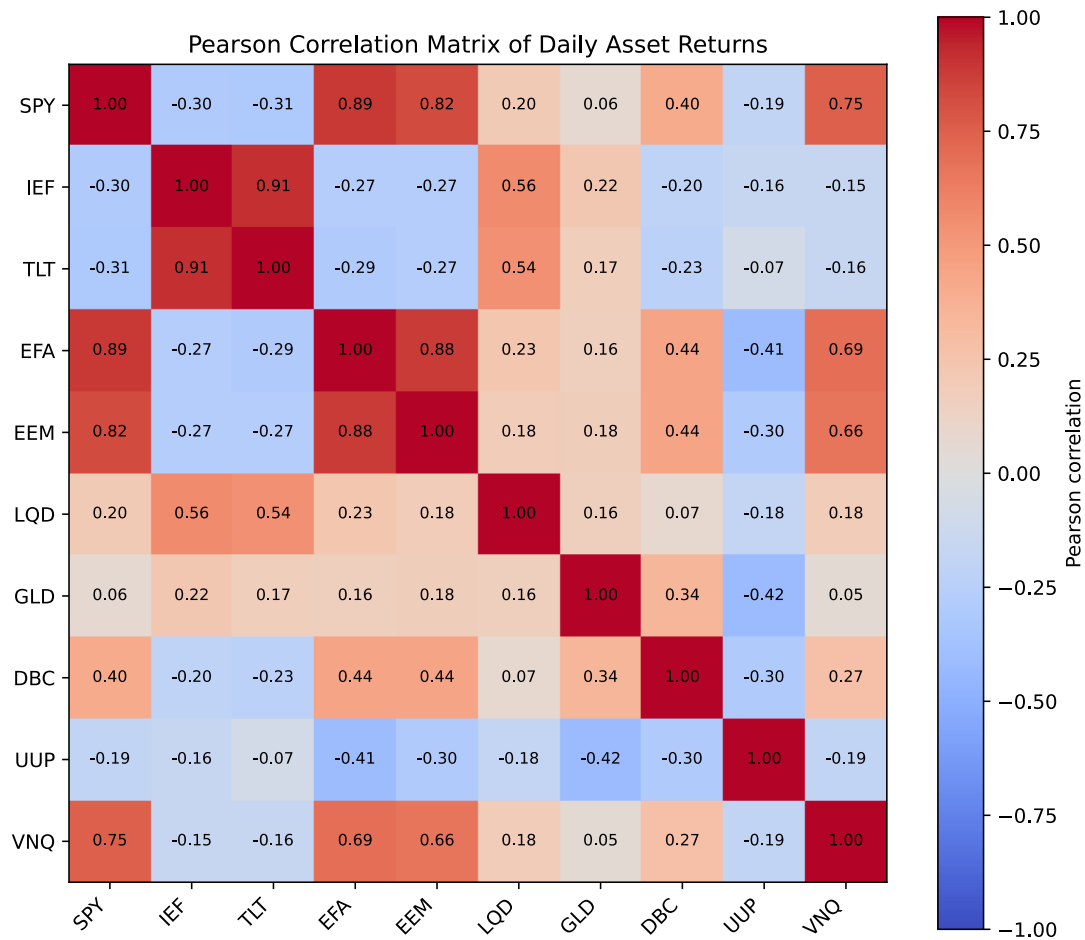


Figure 1: Daily-return correlation matrix of the frozen ETF universe.

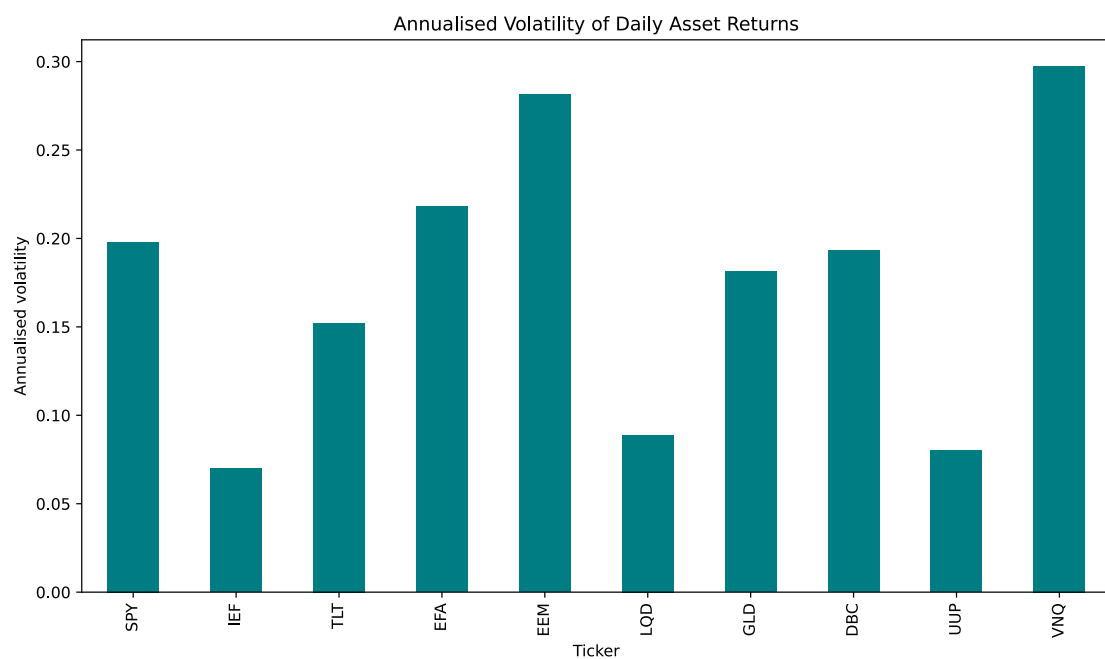


Figure 2: Annualised volatility of daily returns by asset.

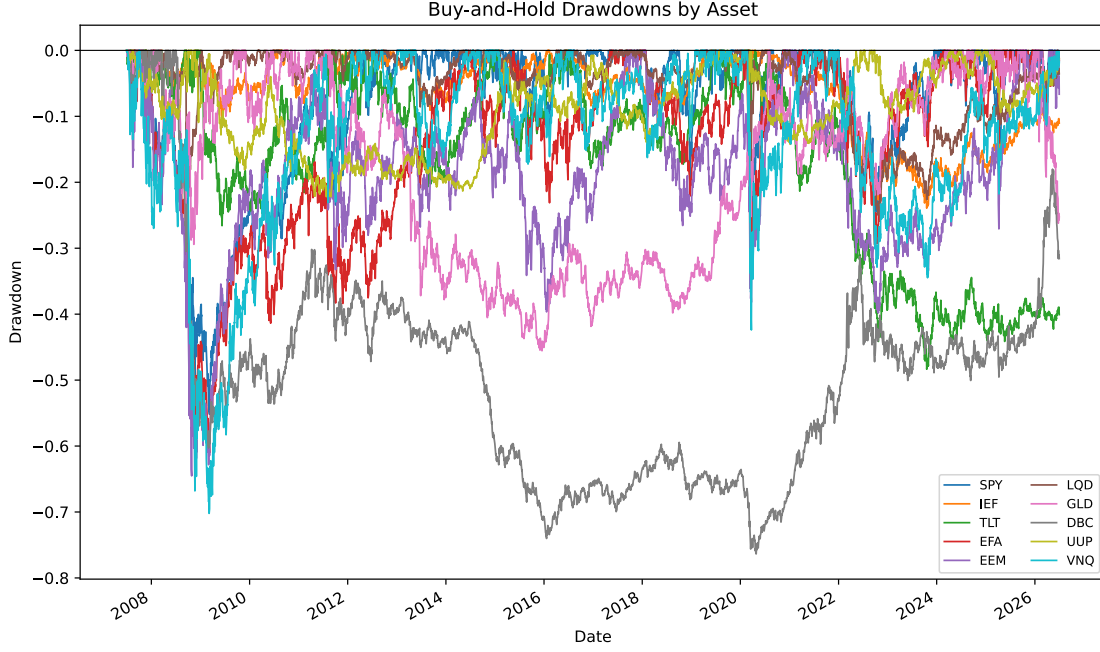


Figure 3: Buy-and-hold drawdowns by asset.

3 Signal definition

For asset i at date t and lookback L , momentum is defined as

$$M_{i,t,L} = \frac{P_{i,t}}{P_{i,t-L}} - 1.$$

The signed signal is

$$s_{i,t} = \begin{cases} +1 & \text{if } M_{i,t,L} > 0, \\ 0 & \text{if } M_{i,t,L} = 0 \text{ or history is insufficient,} \\ -1 & \text{if } M_{i,t,L} < 0. \end{cases}$$

The baseline uses $L = 252$ trading days. Candidate lookbacks are selected only using the validation protocol described below. Targets formed with close data at t are applied by the backtesting engine to the return ending at $t + 1$.

4 Portfolio construction and implementation

4.1 Equal-weight momentum baseline

The baseline target weight is $w_{i,t} = s_{i,t}/10$, rebalanced monthly at 100% gross exposure. Net exposure is signal-dependent and is not constrained to zero.

4.2 Inverse-volatility comparison

For a trailing volatility window W ,

$$\hat{\sigma}_{i,t} = \text{std}(r_{i,t-W+1:t})\sqrt{252}, \quad w_{i,t} \propto \frac{s_{i,t}}{\hat{\sigma}_{i,t}}.$$

Weights are normalised to 100% gross exposure, subject to a 35% individual absolute-weight cap. No portfolio-level volatility target or leverage is used in the initial study.

4.3 Costs and benchmarks

Transaction costs are modelled as one-way proportional costs per unit of turnover. The baseline is 5 bp per asset, with sensitivity at 0, 1, 2, 3, 5, 10, and 20 bp. Benchmarks are the equal-weight long-only universe, SPY buy-and-hold, and individual buy-and-hold asset results.

5 Research design

Table 3: Temporal separation and decision protocol

Period	Dates	Purpose
Train	2008-07-01 to 2015-06-30	Infrastructure, diagnostics, and development
Validation	2015-07-01 to 2019-06-30	Predefined model comparison and selection
Test	2019-07-01 to 2026-07-01	Untouched out-of-sample evaluation

The validation procedure is staged. First, predefined momentum lookbacks are compared under equal-weight monthly construction. Second, the selected lookback is held fixed while equal-weight and inverse-volatility variants are compared. Rebalance frequency is a robustness analysis rather than a parameter chosen solely by maximum Sharpe ratio. No model decision is revised in response to test-period performance.

6 Development-period results

Research status The results below are implementation and development diagnostics only. They do not select a model, alter the frozen design, or constitute an out-of-sample claim.

6.1 Baseline and passive benchmark comparison

Table 4 compares the fixed 252-day, monthly-rebalanced equal-weight momentum baseline with passive alternatives over the development period. All portfolios start with £100,000 and use the same mark-to-market ending convention. The buy-and-hold benchmarks incur only their initial 5 bp entry cost; the momentum strategy also incurs costs when its monthly target weights change.

Table 4: Development-period performance comparison, 1 July 2008 to 30 June 2015

Portfolio	Total	CAGR	Ann. vol.	Sharpe	Max DD	Turnover	Cost
TSM baseline	-2.2%	-0.3%	13.0%	0.04	-29.1%	20.6	1.03%
Equal-weight long-only	31.0%	3.9%	10.0%	0.44	-24.4%	1.0	0.05%
SPY buy-and-hold	85.7%	9.3%	22.5%	0.51	-47.2%	1.0	0.05%

Source: Author calculations from adjusted daily ETF prices. Sharpe ratios use a 0% risk-free rate.

The baseline did not match either passive benchmark on return over this development period. Its maximum drawdown was lower than that of SPY, but higher than that of the diversified long-only portfolio. This is a descriptive diagnostic rather than a decision rule: no parameter or construction change follows from this comparison. With this in mind, the results here are not

providing any evidence toward a possible benefit of the momentum strategy. In fact both the equal-weight and SPY buy-and-hold benchmark enormously outperformed the baseline, with the baseline being the only portfolio to lose money over the development period. While these results do not look promising, they are not sufficient to reject the primary hypothesis. It is also worth noting that none of the portfolios were able to avoid a massive drawdown but also not able to have a Sharpe ratio above 1.0 which is a common threshold for a robust strategy. Thus, no out-of-sample claims can be made from these results and the next stage is to explore the validation period to select a robust configuration across many tunable parameters.

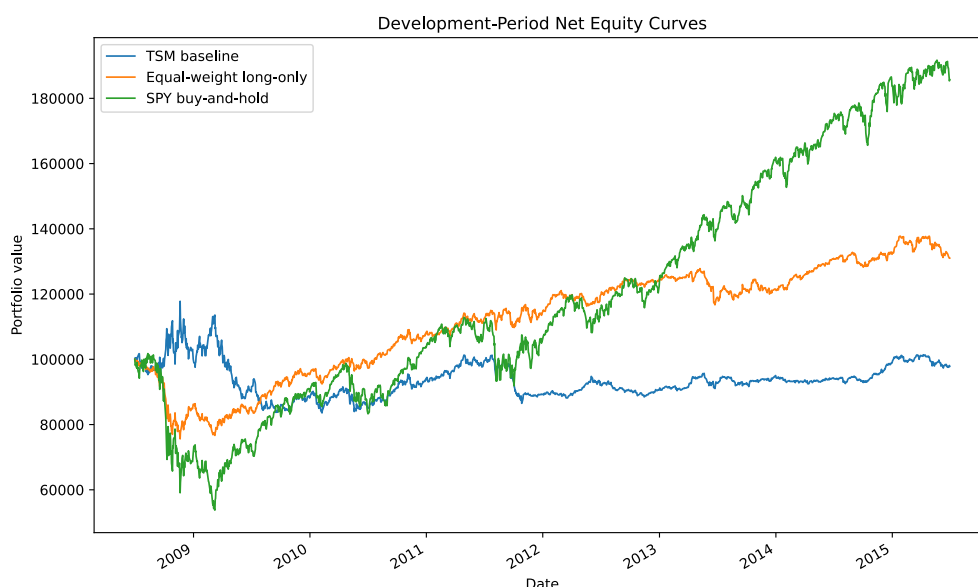


Figure 4: Development-period net equity curves. The comparison is descriptive and precedes validation-period model selection.

7 Validation-period lookback comparison

Research status This is the first stage of model selection: equal absolute weights, monthly rebalancing, and the pre-specified 5 bp cost are held fixed while only the momentum lookback varies. The test period remains untouched.

7.1 Pre-specified lookback family

Table 5 reports the six lookbacks fixed before validation. All rows use the same 1 July 2015 to 28 June 2019 realised trading-date sample; 28 June is the final common trading date within the stated validation boundary. Cumulative cost is shown as a percentage of initial capital.

The 7-day lookback is the only positive configuration, but it is isolated from the rest of the lookback family and has substantially higher turnover and costs. The table therefore does not, by itself, support selecting 7 days solely for its validation Sharpe ratio. A selection decision remains pending the pre-specified cost-sensitivity and stability assessment; it is acceptable for that assessment to conclude that no robust lookback is supported.

Table 5: Validation-period monthly equal-weight momentum comparison

Lookback	Total	CAGR	Ann. vol.	Sharpe	Max DD	Turnover	Cost
7 days	6.8%	1.7%	5.0%	0.35	-5.7%	44.4	2.22%
21 days	-0.6%	-0.2%	5.3%	-0.00	-12.5%	45.6	2.28%
63 days	-9.7%	-2.5%	5.8%	-0.41	-11.9%	25.2	1.26%
126 days	-5.1%	-1.3%	5.6%	-0.21	-13.6%	16.2	0.81%
189 days	-2.0%	-0.5%	5.5%	-0.07	-10.3%	14.8	0.74%
252 days	-6.3%	-1.6%	5.5%	-0.27	-17.2%	13.2	0.66%

Source: Author calculations from adjusted daily ETF prices. Sharpe ratios use a 0% risk-free rate.

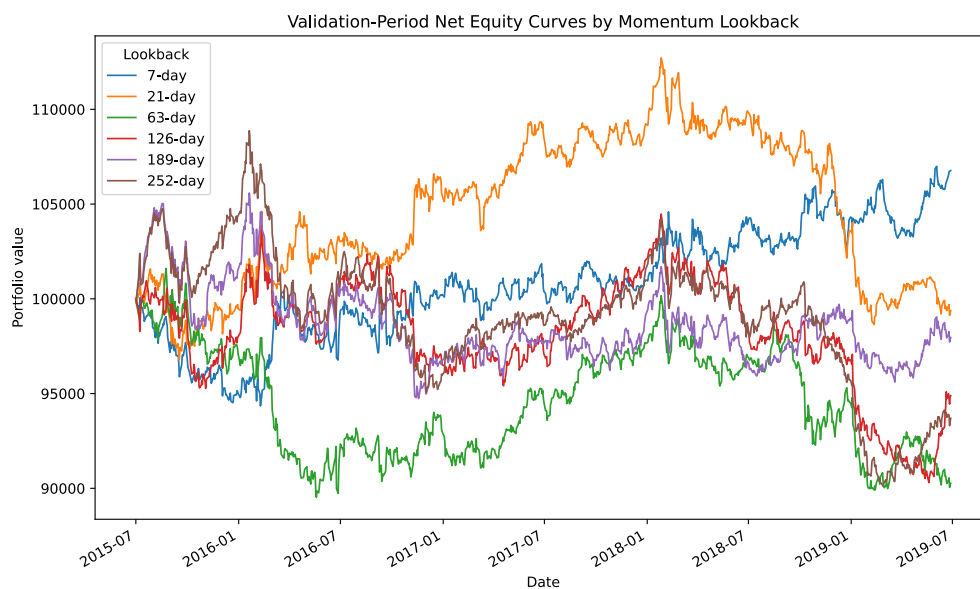


Figure 5: Validation-period net equity curves for the pre-specified monthly equal-weight signal windows.

8 Validation-period cost sensitivity

Research status The pre-specified cost analysis does not support selecting the monthly equal-weight model. The highest Sharpe ratio is 0.46 at zero cost for the isolated 7-day lookback, falling to 0.35 at the 5 bp baseline and 0.02 at 20 bp. No configuration reaches a Sharpe ratio of 0.5 under any cost assumption.

8.1 Pre-specified implementation-cost scenarios

Table 6 varies only the one-way transaction-cost assumption. For each lookback, the monthly target weights and therefore turnover are unchanged across rows; the table is an implementation robustness check rather than an additional parameter search.

Table 6: Validation Sharpe ratio by one-way transaction-cost assumption

Lookback	0 bp	1 bp	2 bp	3 bp	5 bp	10 bp	20 bp
7 days	0.46	0.44	0.42	0.40	0.35	0.24	0.02
21 days	0.11	0.08	0.06	0.04	-0.00	-0.11	-0.32
63 days	-0.36	-0.37	-0.38	-0.39	-0.41	-0.46	-0.57
126 days	-0.17	-0.18	-0.19	-0.19	-0.21	-0.24	-0.32
189 days	-0.03	-0.04	-0.05	-0.05	-0.07	-0.10	-0.17
252 days	-0.24	-0.24	-0.25	-0.25	-0.27	-0.30	-0.36

Source: Author calculations from adjusted daily ETF prices. Sharpe ratios use a 0% risk-free rate.

The 7-day configuration remains the best row throughout the grid, but it is not supported by a broad or high-quality region of lookback performance. Even before costs its Sharpe ratio remains below 0.5, while its higher turnover makes the result increasingly fragile as trading costs rise. The other five lookbacks provide no corroborating evidence. Accordingly, no equal-weight configuration is selected, the inverse-volatility and rebalance-frequency stages are not run, and the test period remains untouched.

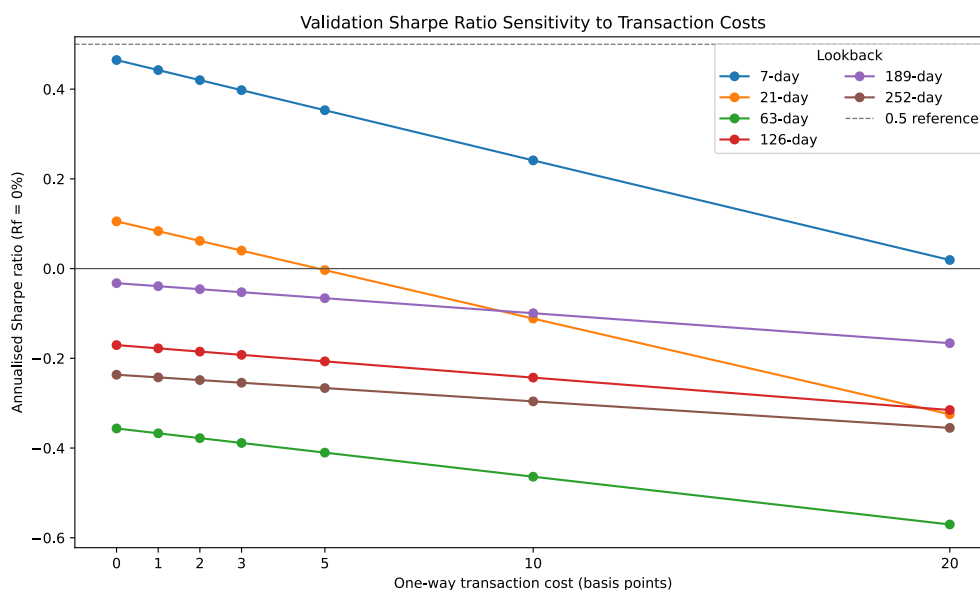


Figure 6: Validation Sharpe-ratio sensitivity to the pre-specified one-way transaction-cost grid. The dashed line is a 0.5 reference, not a fitted decision boundary.

9 Interpretation, limitations, and conclusion

9.1 Current interpretation

The development-period baseline is functioning as specified, but neither the development results nor the equal-weight validation results establish economic value. The pre-specified lookback and cost-sensitivity analysis does not support freezing a robust equal-weight configuration. In accordance with the staged protocol, inverse-volatility construction and rebalancing robustness are not used to search for a rescue result, and the test period remains untouched. This is a negative validation finding for the specified equal-weight time-series-momentum family, not a claim that momentum cannot work under any other universe, implementation, or research design.

9.2 Final conclusion

Research status This research project is complete. No further tests are justified within the frozen design because validation did not identify a robust candidate; the test period remains intentionally unused.

For the specified universe, signal family, equal-weight construction, monthly rebalancing, and transaction-cost assumptions, the primary hypothesis is not supported. The 7-day lookback was the best validation result, but it was isolated from the remainder of the lookback family, achieved a Sharpe ratio of only 0.46 even at zero transaction cost, and fell to 0.35 under the 5 bp baseline. No configuration reached a Sharpe ratio of 0.5 under any pre-specified cost scenario.

The study therefore fails to reject the null hypothesis that this implementation does not provide sufficiently robust economic value after realistic implementation assumptions. No equal-weight configuration is frozen for inverse-volatility sizing, rebalancing robustness, or test-period evaluation. Continuing to vary portfolio construction, timing, or parameters after this validation result would be a post-hoc search rather than a valid continuation of the pre-specified study.